[jair] Editor Decision [20639]

Dear David Herron, Ernesto Jimenez-Ruiz, Tillman Weyde,

We have received all the reviews for the paper you submitted to JAIR, entitled "Logical Reasoning with OWL Knowledge Graphs for Learning and Validating Visual Relationship Detection".

The reviewers agree that the paper proposes a new interesting neruosymbolic method to perform OWL reasoning on top of neural predictions. However, they also identify several difficulties, which preclude publication at this stage. In particular, all the reviewers have pointed at the verbosity of the manuscript and the necessity of some rewriting for the experimental analsysis. You will find the reviewers' comments and suggestions attached. Since JAIR's policy is to accept papers that only require minor revisions, we cannot currently accept the paper. However, we would be happy to see a revised paper on this subject--addressing the reviewers' concerns--published in JAIR. Thus, I strongly encourage you to resubmit the paper.

Please note that papers can be resubmitted to JAIR only once. If you choose to revise the paper, please submit it through our portal, and include a cover letter describing the changes and responding to the reviewers' criticisms.

Thank you for submitting your paper to JAIR.

Regards,

Eleonora Giunchiglia, e.giunchiglia@imperial.ac.uk,

Associate Editor, JAIR

Reviewer A:

This manuscript presents several contributions. More broadly, it substantiates (logical) reasoning in a VRD system by leveraging a KG equipped with an ontology (specified in OWL 2). They apply two strategies, annotation augmentation and prediction validation, to improve a neurosymbolic VRD system.

At a high level, this manuscript seems to have many contributions that are reinforced with significant discussion and many visuals. However, I had significant trouble interpreting the text (verbosity) and the visuals (not well connected to the text). While I do not have in particular complaints with respect to the nature of the research or the technical merit, its _presentation_ obscures contribution (in this reviewer's eyes).

Detailed review, below.

Major Comments:

Sections 2 and 3 are thorough. Section 3 is almost 3 pages, long, however. While I generally appreciate a strong anchoring or comparitive analysis to the state of the art, I think it can be expressed much more succintly. More broadly, the level of detail for this manuscript really makes me feel as if I am a reviewer in the second round, which has tried to cater to every possible comment. In some sense, this leads to a rather thorough discussion. In another, this manuscript is a bit of a slog.

I generally find the figures to be... unhelpful. It's difficult to articulate why, unfortunately. I think there is some combination of a lack of intuition on my part, but sometimes I'm just not sure how clear these are. For example, it's unclear to me how prediction volume can be impacted by the TopN. I also later see that on page 24 (solidly 3 pages after 5.1.1), the baseline is supposed to generate 33 predictions per image. So this must actually be correctness? I've lost the thread, here.

I really don't understand Fig 10? Does this mean that literally every prediction by the baseline is incorrect? That seems incorrect from the previous figures. But I don't see any discussion on this...

What does robust really mean?

Minor Comments:

* Broadly, there is a mismatch between British spellings (e.g., scrutinise) and American convention (e.g., "e.g.,")

> We apply strategy 2 OWL reasoning in our neurosymbolic VRD system during both predicate detector training and prediction.
I think something got mangled in this sentence.

> The formal underpinning of OWL 2 ontologies is based
on the description logic SROIQ
I think more specifically \mathcal{SROIQ^D}?

> And class vrd:Carnivore is declared to be a subclass of class vrd:Mammal
I think this is not strictly true.

* Fig 2 text is pretty difficult to read at a standard zoom (and zooming in shows some strange resolution issues -- stretching?) and hardly in print. There is a lot of space in the figure itself to use a larger font. (As a side note, if you're going to use a flowchart-like syntax for your diagram, you might be interested in using the FvH boxology)

* I'm not sure if Fig 3 is altogether that interesting. It might be better represented as a table, as it stands, given their common heading (and otherwise disjointness), the graphical representation does not seem to add much.

* It would be helpful to distinguish (visually, font-wise) to distinguish "strategy n" (both on page 2 and 13, etc) Similarly for treatments.

* Figure 4 could be improved with a concrete example, similar to figure 1.

> Here, OWL reasoning strategy 1 (annotation
augmentation) is represented by treatment t09 (S_TEU), which turned out to be the best performing configuration of OWL ontology VRD-World, with respect to recall@N, for OWL reasoning strategy 1.
Malformed? Actually several sentences in Section 4.4 should be reviewed (perhaps mangled due to edits?) and adjusted for clarity.

* Fig 6 and other box and whisker plots are missing median lines for the larger spreads. Later, I suppose black dots are used for means.

* I'm really unsure of the relevance of Section 6.2.2 for the rest of the paper.

Recommendation: Reject with encouragement to resubmit

Reviewer C:

The article reports on experiments combining OWL reasoning with neural networks to improve visual relationship detection. Against a baseline of a simple DNN, the authors present evidence for two different integration approaches: 1. augmenting training data with additional inferences, and 2. using the knowledge in the ontology as constraints built in to the loss function.

The topic is interesting and makes a significant contribution to the literature on neuro-symbolic approaches to augment deep neural networks with prior knowledge formalised in logic. However, I have several concerns with the manuscript in its current form:

- Overall the article text is very verbosely written. I would highly recommend editing strongly to reduce repetition of points, which I noticed in several different sections. I would also recommend considering adjusting the tone of the writing to be a bit less discursive and more direct.

- I found myself questioning the choice of baseline (presented in Section 4.1.2). There is no presented justification for the choice of a simple feedforward DNN as baseline classifier for this problem. Given the recent developments in the field, I would have assumed that a natural starting point today would be a pre-trained visual language model as a basis, with added classification layer. If the authors prefer to abstract away from the starting point of visual aspects entirely and only have text object descriptions as input, nevertheless a pre-trained large language model would still be a good starting point. I would highly recommend the authors check the performance of models of this type as an additional baseline.

- In the results, it was not evident to me how many repetitions had been conducted, or what variability in system response could be expected. For the OWL reasoning part, repetitions and variability estimates are of course not needed but for the DNN predictions, assumedly they would be. Ideally the authors should test multiple runs and report variability in the results.

- Although the reported % change in metrics relative to baseline were interesting, I would also like to see tables that just present the results in the usual metrics directly (the actual numbers side by side, not only the % change relative to baseline). This would be more in line with how results are usually presented.

Recommendation: Reject with encouragement to resubmit

Reviewer E:

Summary: this paper presents a neurosymbolic method for recognising relations between entities in images. Given an image – and assuming that the main entities in it have been detected and assigned to a class each – the system predicts relations among them. For example, (person **riding** horse), where the relation “riding” is predicted connecting a person and a horse identified in the image.

The approach combines a “subsymbolic” component - a machine learning (ML) architecture - and a “symbolic” component - an ontology expressed in OWL 2 RL (a rule-like fragment of the Web Ontology Language) and a suitable reasoner. Furthermore, these two components are combined in two different ways, which the paper calls “strategies”. In Strategy 1, the training data (a set of images and RDF triples identifying relations between entities in them) for the ML component is enriched with additional triples using the OWL reasoner to compute new inferences, i.e., augmenting the training

data with new triples logically implied by the combination of this data and the ontology. In Strategy 2, the OWL ontology and reasoner are used to semantically validate the predictions of the ML component. This is done both in training and in testing: in training, the predicted triples by the ML system are verified or disproved by the OWL reasoner (this checks whether the predicted triples result in a contradiction, when considering the input data and the given ontology); this informs the loss function, which penalises (soft constraint) the prediction of inconsistent triples. In testing, triples found to be inconsistent are simply removed (hard constraint).

The paper performs extensive experiments over a novel benchmark previously introduced by the authors: NeSy4VRD, consisting of images and annotations of relations between entities detected in them. We note that the vocabulary of the examples in the NeSy4VRD benchmark and the OWL ontology used in this approach are essentially matching. The results of the experiments show that both Strategy 1 and Strategy 2 (applied separately, or applied together) result in an increase in the performance and recall of predictions, with few false positives – in fact several “false positives” turn out to be plausible predictions in the image (as determined by manual evaluation) which simply were not present in the annotated dataset. Furthermore, the Strategies accelerate the training of the system.

Strengths:

-Interesting and timely neurosymbolic system. The work provides a concrete demonstration that OWL reasoning can be paired with deep learning to produce measurable gains in performance, superior to that of either system individually.

-Thorough testing of the different types of OWL axioms and their interactions. The experimental design considers many different copies of the systems, which differ in which types of axioms (among subsumption, inverse, equivalent relations) are allowed to participate in inferences. This allows for a detailed understanding of the interactions between these types of axioms, in their contribution to the overall system.

-Clear writing in methodology and motivation. The overall pipeline and the specific reasoning strategies are well explained, making the technical novelty accessible.

-Strong literature review. The paper covers well existing relating literature: it includes many well-known works that are connected to this area, and it is easy to understand the position of the current paper in it.

Weaknesses:

-The Results section is somewhat wordy and hard to follow. The key findings are sometimes buried in long, detailed sentences that make it difficult to grasp the main conclusions. A clearer description of the key results, with less step-by-step narration, would strengthen the message and free up some space for deeper analysis (see point below).

-Experimental Results are not analysed in depth. Related to the point above: the experimental section shows several unexpected behaviours (such as inhibition of predictions when using strategy 1 with inverse properties, or the fact that making the soft constraints in strategy 2 harder can reduce the system's performance), but the section mostly describes them, rather than performing further experiments to understand their cause. Some of these are set for Future Work, but it would have been interesting to see that here.

-Concerns about significance of the results: the approach relies on a highly curated benchmark and a bespoke ontology. To apply the method in a different setting, one would need to produce another set of annotated images and a new ontology, and it is unclear whether the observed behaviours would generalise (especially because the paper covers only 1 use case).

OVERALL ASSESSMENT: the paper offers a very interesting a timely example of a neurosymbolic approach to a task. The methodology is clearly described and sound, and its effectiveness is amply demonstrated. The weaknesses highlighted above do not compromise the core contribution and its value, so I recommend acceptance. The detailed comments below are aimed at improving a potential camera-ready version of the paper.

DETAILED COMMENTS

Page 2: on the first mention of "*predicate detection subtask*." I recommend explaining this in one sentence and how it connects to relationship detection. I first believed they were essentially the same thing, then the start of section 4.1.1 explained the difference very clearly. I'd suggest mentioning this much earlier, to facilitate reading the introduction.

Page 2: a general comment on the paper's motivation: one of the main motivations of the paper is that the "symbolic" part is based on OWL-based reasoning specifically, in a way "fulfilling the promises" of this language as a vehicle for semantic technologies. However, the actual implementation does not seem to use a lot of OWL: the ontology is all within the OWL 2 RL profile – and not just that, but a fragment of it. Other than Transitivity, it seems most features are very close to just RDFS.

Page 6: I am very pleased to see that the VRD data was restored. This is a great contribution (from the authors, though not a direct contribution in this paper, if I understand correctly). How was the restoration performed? Was this done manually? It might be worth mentioning.

Page 7: *"We inject further common sense, symbolic background knowledge into VRD-World by describing these object properties (these counterparts of NeSy4VRD predicates)"* Can you clarify how you do this?

Page 7: Did you introduce additional VRD relations when writing the ontology, or did you restrict your signature to that of the NeSy4VRD annotations? Intermediate relations might be useful (since reasoning is performed using essentially Datalog semantics, which allow for recursion).

Page 11: could you say just a bit more about the object detection part? Although it's not particularly relevant to the paper, it might be nice to include a bit more of information. The paper refers to (Herron, 2025) but perhaps you could explain, for example, how much noise needed to be corrected?

Page 12: can you say more about what other geometric and spatial features derived from the bounding boxes you added, and how much this helped the performance of the model?

Page 13: is it ok to perform model selection using test data? I would expect to consider only validation data for model selection. But it seems here that the checkpoint with largest recall is chosen, where recall is evaluated on test data.

Page 13: "*The axioms of OWL ontology VRD-World use OWL constructs that correspond to the five subcategories of link inference semantics highlighted in Figure 3. We arrange to activate different subsets (combinations) of these five subcategories of link inference semantics so that we can study their differential effects on the subsymbolic learning of our predicate prediction neural network (PPNN).*" This was very confusing and difficult to read. I suggest clarifying this. It took me several re-reads to understand this.

Page 14: any intuition for why you introduced a class to represent the completement of classes, instead of negation? Is it because it is not fully supported by OWL 2 RL?

Page 15: why does the call to the reasoner not appear in Figure 4?

Page 17: can you explain more why you use soft penalties during training, instead of harder ones? Is this because of possible noise in the inferences? I later noticed this question is left for future work – it would be interesting, however, to hear some hypotheses for why this happens.

Page 17 “*by the symbolic reasoning binary classify”* is there some typo here?

Page 18: I’m confused by this: “*every visual relationship is an instance of a type, and once the type is classified as valid or invalid (triggered by an encounter with a first instance), so are all further instances.”* Aren’t types and relationships different things? What does it mean for a type to be valid?

Page 18: “*The requirement for a match between two bounding boxes is that their IoU…”* What is this requirement? Is it a standard requirement or one defined by you? If defined by you, why do you need it? I’d expect boxes match exactly in this experiment. The end of the paragraph also suggests that this is the case.

Page 18: I don’t understand the justification for the formula in 4.5.3

Page 20: The observation that there are “false false positives” is a very interesting piece of analysis. The adjusted precision measure seems one of the most important metrics, but it can only be measured in a rather limited way. Have you looked into automating the evaluation via other image recognition systems, and do some stochastic verification of that system’s outputs? This could offer a much more extensive evaluation, even if potentially less reliable.

Recommendation: Accept with minor revisions
